ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY MODULE 3

UNIT: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: 0612 451 07A

PRACTICAL ASSESSMENT – OBSERVATION CHECKLIST

64% 72%

Time 4 hours

A. CANDIDATE AND ASSESSOR DETAILS

Particular	Details
Candidate Name	JOY MWIRIGI
Candidate Registration No.	698
Assessment Date	8/10/2026
Start Time	9:00am
Assessor Name	James
Assessment Venue	computer Lab

B. ASSESSMENT INFORMATION

Assessment Decision

The candidate shall demonstrate the ability to plan, install, configure, test and troubleshoot a small wired computer network using the equipment provided.

The assessor shall observe the candidate performing the practical activities and award marks based on demonstrated competence.

C. MATERIALS, TOOLS AND EQUIPMENT / CUTTING LIST

No.	Item / Material / Equipment	Quantity	Provided By
1	Desktop computers	3	Centre
2	USB printer with USB cable	1	Centre
3	8-port or 24-port switch	1	Centre
4	Router / wireless router with internet access	1	Centre
5	Cat5e/Cat6 UTP straight-through patch cables	5	Centre
6	RJ45 connectors	As required	Centre
7	Crimping tool	1	Centre
8	Cable stripper	1	Centre
9	Cable tester	1	Centre
10	Diagram sheets	As required	Centre
11	Network labels	As required	Centre
12	Pen/pencil/ruler	As required	Candidate

D. PRACTICAL OBSERVATION CHECKLIST – 50 MARKS

Instructions to Assessor:
Observe the candidate while performing each task. Award marks only for work demonstrated by the candidate. Use the comments column to record evidence, errors, corrections or observations.

No.	Item to be evaluated	Allocated Marks	Candidate Score	Assessor Comments
TASK A – PHYSICAL NETWORK DIAGRAM				
1	Draws a neat physical network diagram showing HR-PC, ACC-PC, REC-PC, switch, router, USB printer and ISP/Internet cloud.	2	✓	-Neat diagram -Indicated -Printer indicated -Cloud included
2	Correctly connects the three computers to the	2		

	central switch and connects the switch to the router.		01	Interconnection done
3	Shows the USB connection between the printer and REC-PC and identifies REC-PC as the print host.	2	02	USB connection shown
TASK A SUBTOTAL		**6**	06	
TASK B – LOGICAL TOPOLOGY DESIGN				
4	Draws the logical topology showing IP addresses, subnet mask, default gateway and internet path.	2	02	Logical topology drawn
5	Correctly shows how HR-PC and ACC-PC access the shared printer through REC-PC.	1	01	demonstrated
6	Identifies the physical topology as star and the logical topology appropriately.	1	01	topology identified
7	Gives valid reasons for selecting the topology, such as easy fault isolation, expansion and centralised management.	1	01	
TASK B SUBTOTAL		**5**	04	
TASK C – ADDRESSING PLAN				
8	Identifies the network as 192.168.10.0/24 and gives the correct subnet mask.	1	00	Not identified
9	Assigns 192.168.10.1 as the router LAN address and uses a static address.	1	00	
10	Assigns unique IP addresses to HR-PC, ACC-PC and REC-PC within the correct subnet.	3	03	unique IP addresses assigned

11	Assigns the correct default gateway to the computers.	2	02	Incorrect gateway
12	Includes the switch management address where applicable and provides a documented growth/reserved address range.	1	01	
TASK C SUBTOTAL		**8**	03	
TASK D – INSTALLATION AND CONFIGURATION				
13	Connects the network devices correctly using Cat5e/Cat6 straight-through cables and labels cables/ports.	2	02	devices connected using straight cable
14	Correctly demonstrates preparation/testing of a straight-through cable where required and obtains a successful tester result.	1	01	demonstrated
15	Configures the router LAN IP, hostname and management password correctly.	2	02	Configured
16	Configures the required DHCP/NAT/internet-sharing settings where applicable.	1	01	DHCP configured
17	Renames the computers as HR-PC, ACC-PC and REC-PC and places them in the BRIGHTPATH workgroup.	3	03	Renamed the computer
18	Correctly configures the IP address, subnet mask and default gateway on the computers.	1	01	Configured
19	Connects the USB printer to REC-PC, installs the driver and enables printer sharing.	1	01	Connects host to printer

20	Adds the shared printer to HR-PC and ACC-PC and demonstrates successful printing.	1	1	
21	Creates and shares CompanyShare so that all three computers can access it.	1	1	Created & shared the
22	Creates FinanceHR and configures access so that HR and Accounts can access it while Reception is denied.	1	0	not configured
TASK D SUBTOTAL		**14**	13	
TASK E – TESTING AND VERIFICATION				
23	Tests connectivity between HR-PC and ACC-PC using ping and obtains successful replies.	1	1	Tested
24	Tests connectivity between ACC-PC and REC-PC using ping and obtains successful replies.	1	1	Ping sent
25	Tests connectivity between REC-PC and HR-PC using ping and obtains successful replies.	1	1	Ping sent
26	Tests connectivity from each computer to the default gateway.	1	1	Tested
27	Verifies that the shared printer is visible from HR-PC and ACC-PC.	1	1	Verified printer is visible
28	Verifies internet/ISP connectivity from all three computers.	2	2	Internet connect verified
29	Performs a tracert test and correctly identifies the first hop as the default gateway.	1	0	did not attempt.

30	Prints a test page from the required computers and records the result.	1	1	printed
31	Tests access to CompanyShare from all three computers.	1	1	Tests access
32	Demonstrates that Reception is denied access to FinanceHR and records the result.	1	0	not demonstrated
TASK E SUBTOTAL		**11**	09	
TASK F – TROUBLESHOOTING				
33	Identifies the network symptom and follows a systematic troubleshooting procedure.	2	0	
34	Uses appropriate diagnostic tools/commands such as ipconfig, ping, tracert and physical inspection.	1	1	
35	Correctly identifies the cause of the introduced fault.	1	0	
36	Corrects the fault without causing unnecessary changes to the network.	1	0	
37	Re-tests the affected connection/service and confirms that the network is working.	1	0	
TASK F SUBTOTAL		**6**	~~1~~ 0	
TOTAL PRACTICAL OBSERVATION MARKS		**50**	0	

E. PRODUCT CHECKLIST – 10 MARKS

Instructions to Assessor:
Inspect the completed network installation and other products produced by the candidate. Award marks according to the quality and functionality of the final products.

No.	Product / Expected Outcome	Allocated Marks	Candidate Score	Assessor Comments
1	Neat and correctly labelled physical network diagram showing all required devices, connections, ports and cable types.	2	2	
2	Complete and accurate addressing table using the 192.168.10.0/24 network, including gateway and reserved growth range.	2	2	
3	Functional wired star network with correct device connections and working network links.	1	1	
4	Functional shared USB printer accessible to the required computers through REC-PC.	1	1	
5	Functional CompanyShare folder accessible by all three computers.	1	1	
6	Functional FinanceHR folder with access restricted to HR and Accounts and Reception denied.	1	0	
7	Completed network test log with accurate Pass/Fail results and supporting test evidence.	1	0	
8	Neat cable/port labelling and organised network installation.	1	0	
TOTAL PRODUCT MARKS		**10**	05	

CANDIDATE FEEDBACK

Candidate Feedback on the Assessment

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.....Fair.....

Candidate's Strengths

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Candidate's Areas for Further Improvement

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MARK SUMMARY

Assessment Component	Maximum Marks	Candidate Score
Practical Observation Checklist	50	
Product Checklist	10	05
TOTAL	60	

Percentage: _____%

L. COMPETENCE / DECISION OUTCOME

Decision	Tick
Competent	☑
Not Yet Competent	☐
Re-assessment Required	☐

Particular	Name / Signature	Date
Candidate Name	Jay Mwiriji	
Candidate Signature		
Assessor Name	James	
Assessor Signature		
Internal Verifier / Moderator		

ASSESSOR DECLARATION

I confirm that I have observed the candidate performing the practical activities and that the marks awarded are based on the candidate's demonstrated performance and the products presented during the assessment.

Assessor Name: James

Assessor Signature: [signature]

Date: 08/1/2026